AI FAQ & Customer Support Automation

REQUIREMENT ANALYSIS DOCUMENT

PROJECT OBJECTIVE

Develop a secure, scalable REST API combining customer support ticketing and FAQ management with Google Gemini AI for automated query resolution, ticket prioritization, and support insights.

Functional Requirements

- User and Support Agent registration/login with JWT.
- Protected profile and role-based access control.
- Create, view, update, and delete FAQs & tickets.
- Search FAQs and tickets by title, category, status, or date.
- Generate AI-powered answers for customer queries.
- Generate AI support insights and auto-replies.
- Return structured JSON & consistent errors.

Data Requirements

- **User:** `_id`, `name`, unique `email`, `hashed password`, `role`, `timestamps`.
- **KnowledgeBase:** `_id`, `title`, `category`, `content`, `tags`, `timestamps`.
- **Ticket:** `_id`, user ref, `subject`, `category`, `status`, `priority`, `description`, agent ref, `timestamps`.
- **Categories:** General, Billing, Technical Support, Account Access, Product Inquiry, Refunds.

SECURITY & NON-FUNCTIONAL REQUIREMENTS

- JWT protection for protected routes; `bcrypt.js` password hashing.
- Mongoose schema validation; environment variables for sensitive API keys/secrets.
- Modular MVC design pattern; centralized structured JSON error handling.
- Maintainable and intuitive RESTful API surface.

Software Requirements

- **OS:** Windows 10/11, macOS, or Linux
- **Runtime/Env:** Node.js v16+, npm v8+
- **Framework/DB:** Express.js, MongoDB
- **Tools:** VS Code, Postman / Thunder Client
- **Packages:** `express`, `mongoose`, `dotenv`, `bcryptjs`, `jsonwebtoken`, `cors`, `nodemon`

Hardware Requirements

- **Processor:** Intel i5 8th Gen+ / Ryzen 5 equivalent
- **Memory:** 8 GB RAM minimum (16 GB recommended)
- **Storage:** 1 GB available disk workspace